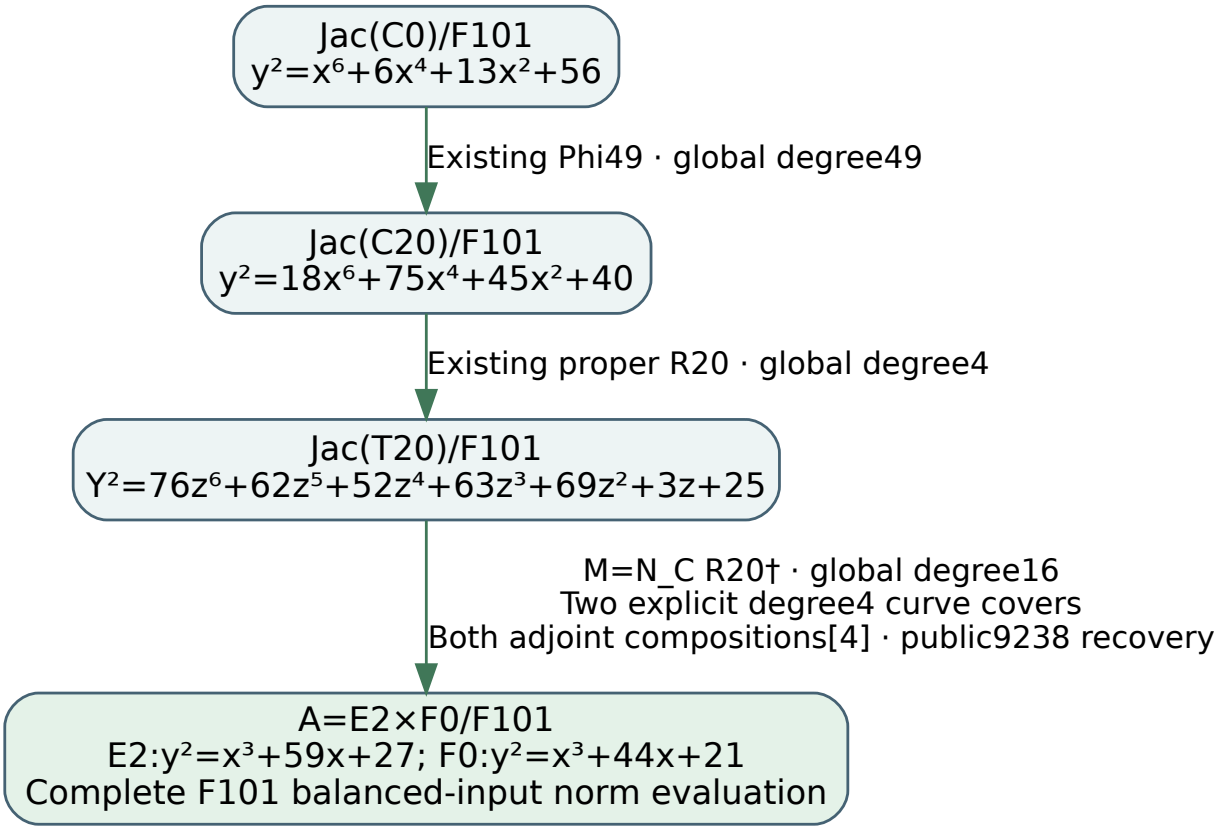
Prime-field T20 degree-four elliptic norms

2026-11-12 · full43/85436f · written proofbfa422 · unsigned

The actual M=N_C20 R20† is now evaluated on every12317 rational balanced T20 input using F101 arithmetic and no input-support roots. Both actual adjoint compositions are[4],with public9238 recovery,group/Frobenius and all109+113 scalar images/exact orders checked. [pcc:unverified]



For raw T20 Y²=f(z),the two covers have X=n/q²,Y_E=Y·m/q³:

E2(59,27)	q=z²+38; n=10z⁴+54z²+52z+53; m=2z³+55z²+47z+44; infinity+=(10,100).
F0(44,21)	q=z²+46; n=71z⁴+96z³+92z²+49; m=5z³+53z²+17z+87; infinity+=(71,48).

Y=50·y_monic fixes the raw leading76 by a square coordinate isomorphism. The two curve covers have global degree4;the induced Jacobian norm pair has degree16. Constructor setup still uses and must charge the existing F101¹² factory. No timing/cost result is claimed. [pcc:unverified]

Complete input domain and controls

Complete rational input population · full43/85436f

Unit: classes · n=12317 · disjoint degree categories · exact counts

Wrong adjoint sign:12316 different /1 equal (identity)

Repeated118 · nonsplit5116 · pole inputs2 per cover

Additional controls overlap;not sampling estimates or cost observations.

Full43 retains every old41 test,reference/legacy guards,fmt/strictClippy, all628 input hashes and byte-identical geometry6141/kernel24475 JSONs. Static1bbfec,geometry58b615 and whole90601c are separately retained. Full43 has original start/finish/exit0/afterguards;its whole18-file tree is one57652-byte standardtar.xz with member/raw/archive hashes and exact roundtrip. Source freeze ended only after all receipts. [pcc:unverified]

Geometry controls include120 rational point classes,six branches,eight actual pole points,both signed infinities,128 actual repeated Mumford doubles,Frobenius/coherent-translation invariance and six malformed mutants. Actual16 kernel classes/all256 sums/all killed4/exactly4 killed2 check the(4,4) type;degree16 alone does not imply that type. [pcc:unverified]

The input adapter checks canonical F101/model tags,monic degree≤2, ordinate/infinity counts and the curve relation. Every actual class converts and roundtrips through an explicit rho50 Chart. Runtime evaluates the two coordinates in F101[z]/u with existing invmod and symmetric elliptic sums,including coincident/opposite/repeated jets. It removes only actual completed pole support,with multiplicity,and retains original degree in(2n+degree−2)·image(infinity+). [pcc:unverified]

The runtime type contains Zp,Vec(u64),Curve(u64),Pt(u64);it accepts no extension context,root cache,index or logarithm label. Existing field/ poly/elliptic/Cantor engines are reused. Extension routines belong to setup/ reference geometry controls;no arbitrary $F101^{12}$ Jacobian input domain is inferred. [pcc:unverified]

The wrong global adjoint sign differs on12316 classes and agrees only at identity. Each cover has238 applicable missing-infinity-correction mismatches and two actual pole inputs. Coherent cover translation is a positive norm-invariance control,not a mislabeled failed map. [pcc:unverified]

Exact proofs, comparison and evidence boundary

Fourteen faithful Lean statements include both actual characteristic101 curve equations,explicit Bezout witnesses for numerator/q coprimality, quadratic/linear/translation algebra,conditional norm4/recovery and constants. Final full40module gate52bbc7 audits238 compiled/212 written/ 0trusted/all44 hashes unchanged. Its entire171-file tree is one39344-byte standardtar.xz with complete member/raw/archive hashes/exact roundtrip. [pcc:unverified]

Explicit coprimality and degree4 of n/q^2 give degree4 of the curve covers by the function-field tower. The actual proper reverse fibre,derived adjoint sign and two-torsion translation descend the functions. Constants cancel on degree-zero divisors,so these covers induce N_C20 R20†. Picard/polarization/descent/projective completion and elliptic-jet/group bridges remain written/native;the Lean statements do not formalize the complete Jacobian construction. [pcc:unverified]

Algebra02/9214e7 passes24 tests(old23 retained),eleven outputs,all ten old payloads byte-identical,24 guards and the new mutation rejection. IDC1h6de045a64b58c1c8 checks a universal linear-image ordinate identity over fixed$2^{61}-1$;the characteristic101 equations have separate symbolic Lean proofs. Candidate01 and full40/236 gate4a3e09 stay immutable after proof strengthening. Identity ID is stable;the ProofRef payload changes. [pcc:unverified]

All12317 actual C0 classes satisfy M R20 Phi49= [2]diag(alpha49,id)N_C0. Public normalization2 on the odd rational groups recovers the existing direct norm/isogeny bridge. It is not a global inverse isogeny. No additional solving advantage follows from this comparison. Complete setup/conversion/transfer/solve/ recovery/verification costs and matched external references remain unmeasured. [pcc:unverified]

The first proof-binding batch5176f3 returned not_covered2 because Lean was absent from that exec’s PATH,despite the standalone10-theorem gate passing. The partial c20986 claim/source/verdict/error are retained; correct-PATH12 bindings and the two new degree bindings pass separately. No failed or partial batch is promoted;no frozen report is overwritten. [pcc:unverified]

Local Linuxx86_64/Rust1.99.1/Lean4.35.0 correctness/development,no benchmark isolation. Own new-root CI is next;old consumer8072 all4CIb3c8f1 is settled at its own41 scope. Source-linked REPORT.md lists every custody,exact statement binding,old/new certificate and editable figure/data/renderer. Canonical IC graphs/ catalog rows stay unchanged:no measured cost gain. [pcc:unverified]

Sources:BD0902.3480§2 Lemma2.6/§7 Lemma7.5/§8;Galbraith8.3.8/8.3.10; 720a59/b24be8/DirectNorms/ ClosedFibres/native engines;external CT50 primary dd4077d1,its scripts not adopted. Actual KB127 receipts retained,no novelty inference. Protocolc05a75 preceded code;next distinct consumer pin/replay and independent F101 checks after commit. Scientific standing unsigned. [pcc:unverified]